



Automation Level 2 Badge

Participant Resource Sheet

Name: _____

Email used for badges: _____

Topic	Task	Mentor Signature	Date
Pretest (Start)	Complete the Badge 2 PreTest. Step 1		
Java Foundations	Complete the reference and records exercises. Explain validation, ownership, and callbacks. Step 2		
Project Setup	Create the Badge 2 repository with both drivetrain projects from AdvantageKit v26.0.2 Step 3		
Hardware Setup	Compare FX, FXS, and Nova; configure the hardware and demonstrate bounded operation with mentor supervision. Step 4		
IO and Generated Code	Create Badge2Bench within the Badge 2 repository with PhoenixV6 and ThriftyLib; demonstrate mode selection, replay, Lombok builders, and generated logging inputs. Step 5		
Testing Fundamentals	Pass configuration and controller tests. Explain a failed assertion, inheritance, and method overriding. Step 6		
One Complete Mechanism	Build the “Roller” for our FRC 2027 Mechanism library. This includes support for Talon FX, Talon FXS, and ThriftyNova, in simulation and on the supplied physical fixture. Verify conversions, bindings, stop, and disable. Step 7		
Mechanism Library	Complete roller, flywheel, pivot, and elevator simulations, required tests, saved baselines, and the CTRE/Thrifty profile comparison. Step 8		
Mechanism Composition	Demonstrate the elevator assembly and feed-readiness exercise from Riptide; test interlocks and explain command ownership. Step 9		
Pose and Frames	Complete the frame, transform, odometry, and vision-math checks; explain residuals and uncertainty. Step 10		

Path Following	Complete the guided exercises on PathPlanner and BLine. Include mechanism actions, controller integration, and logs. Step 11		
Observations and Vision	Run two simulated cameras with PhotonVision. Demonstrate filtering, logging, replay, queue ownership, and update order. Step 12		
Robot State	Demonstrate shared pose, derived facts, and tele-op triggers while preserving subsystem ownership. Step 13		
Integration Capstone	Demonstrate selectable autonomous routines with mechanisms and vision, safe stopping, saved logs, and successful replay. Step 14		
Post-Test and Final Review	Score at least 90%, review the practical evidence, and answer the five final questions. Step 15		